

SOPHIA PHELPS

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EDUCATION

University of California, Irvine Physics, Ph.D.	Sept. 2026 – June 2031 (Expected) Irvine, CA
University of California, Los Angeles Physics, B.S.	Sept. 2022 – June 2026 Los Angeles, CA

RESEARCH AND WORK EXPERIENCE

University of California, Irvine NSF Graduate Research Fellow (Advisor: Prof. Paul Robertson)	Sept. 2026 – Present Irvine, CA
Jet Propulsion Laboratory, California Institute of Technology Student Research Intern, SURF@JPL (Mentor: Dr. Gautam Vasisht)	June 2025 – Present Pasadena, CA
Regan Research Group, University of California, Los Angeles Undergraduate Researcher (Advisor: Prof. Chris Regan)	March 2025 – Jan. 2026 Los Angeles, CA
UCLA Learning Assistant Program Education Research Intern (Advisor: Prof. Shanna Shaked)	Jan. 2026 – June 2026 Los Angeles, CA
StudySTEM Founder	2021 – Present Los Angeles, CA

AWARDS, GRANTS, AND HONORS

Total: \$170,600	
NSF Graduate Research Fellowship, National Science Foundation	2026
Summer Undergraduate Research Fellowship, California Institute of Technology	2025
E. Lee Kinsey Memorial Scholarship, University of California, Los Angeles	2025

PUBLICATIONS

- Phelps, S., et al. “Equivalent Width Diagnostics of Solar Magnetic Activity.” *UCLA Undergraduate Science Journal*, published May 28, 2026.

PROGRAMMING AND TECHNICAL SKILLS

- **Programming:** Python (NumPy, SciPy, Pandas, Matplotlib, Astropy, barycorrpy, scikit-learn, PyTorch, and Keras); MATLAB; C++.
- **Scientific computing and tools:** Git, Linux/Bash, Jupyter, VS Code, LaTeX, and FITS.

TEACHING, SERVICE, AND OUTREACH

University of California, Irvine Graduate Teaching Assistant <i>Course:</i> Physics 7C (Fall 2026).	Sept. 2026 – Present Irvine, CA
UCLA Learning Assistant Program Pedagogy Head Learning Assistant; Course Coordinator; Learning Assistant <i>Courses:</i> Physics 1C (Spring 2024); Physics 5A (Fall 2024 and Spring 2026); Math 31A (Winter 2026).	April 2024 – June 2026 Los Angeles, CA
MIPA, University of California, Los Angeles Upper-Division Mentor	Oct. 2025 – June 2026 Los Angeles, CA
Upsilon Lab, University of California, Los Angeles External Vice President	Sept. 2024 – June 2026 Los Angeles, CA

PRESENTATIONS

Oral Presentations

- “Near-Infrared Solar Lines and Magnetic Activity for Improved Radial Velocity Precision in Sun-like Stars.” JPL SIP Final Presentation, Pasadena, CA, August 2026.
- “Near-Infrared Solar Line Diagnostics of Magnetic Activity for Improved Radial Velocity Precision.” SURF@JPL Final Presentation, Pasadena, CA, August 2025.

Posters

- “Near-Infrared Solar Line Diagnostics of Magnetic Activity for Improved Radial Velocity Precision.” Southern California Conference for Undergraduate Research, CSU Channel Islands, Nov. 2025.